A REMARK ON A PAPER OF EBENFELT AND GANGULY

XIAOSHAN LI

I recently noticed a 37-page arXiv preprint by Ebenfelt and Ganguly [EG26] posted on October 3 that substantially overlaps in results and proofs with our 11-page preprint [LL26] posted on September 21, with Taishun Liu. These notes compare the two papers.

1. Mathematical comparison

Both papers prove that a Bergman metric with constant scalar curvature (CSC) is Kähler–Einstein (KE). [EG26] also show that an extremal Bergman metric has constant scalar curvature, using the fact that a holomorphic function vanishing on the boundary vanishes identically. This additional step occupies less than two pages; the others deal with the problem treated in [LL26].

Despite some differences in technical formulation, both proofs follow essentially the same two-step strategy and mechanism:

(1). Show that the Bergman invariant function vanishes to infinite order along a strongly pseudoconvex boundary patch.

(2). Establish boundary unique continuation for the ACH operator Δ_g.

The main part of (1) was established in [HL26], posted in May 31, 2026: Using Martin's fundamental formula from 2021, [HL26] showed that CSC implies local sphericity near a strongly pseudoconvex boundary point. The explicit Bergman kernel formula for the ball then gives, in a few lines, $J_\Omega = \frac{(n+1)^n\pi^n}{n!} + O(\rho^{n+2})$. Applying CSC again, together with Engliš's formula and the comparison argument in [HL26], yields infinite-order vanishing of $\log J_\Omega - \log \frac{(n+1)^n\pi^n}{n!}$ along the boundary. This argument occupies approximately 2.5 pages in [LL26].

The corresponding argument in [EG26] is essentially the same, apart from technical differences and the rederivation of results already available in the literature, including the crucial sphericity. Although [HL26] was sent to the first author of [EG26] by us before its arXiv posting in May 2026, neither it nor its later inclusion as an appendix to [HHL26] is mentioned in any place of [EG26].

The authors of [EG26] credited the basic formula mentioned above to Ebenfelt–Xiao [EX24], although it was originally due to Martin, whose paper is not listed in the references of [EG26]. I became aware of [EX24] while writing these notes and found that it does state Martin's result, with an acknowledgment of Martin's thesis.[EX24] also observed that sphericity improves the vanishing order of $\log J_\Omega - \log \frac{(n+1)^n\pi^n}{n!}$ to $n+2$, a very small part of our proof for (1) that we will acknowledge in a revision of [LL26]. The remainder of [EG26]'s argument is again more or less the same as in [LL26] with difference at technical level.

(2) concerns boundary unique continuation (UCP), the fundamental component of both papers. The presentations and many of the arguments follow essentially the same approach and rely on the same mechanism, with some technical differences;

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for example, certain technical lemmas are stated in slightly more general settings in [EG26]. In [LL26], this part occupies approximately four pages: the authors invoke existing results, verify the required hypotheses, and apply the BH theorem directly. In [EG26], this material appears in Section 6 and occupies approximately fourteen pages.

2. A brief history of [LL26]

We briefly recall the development of [LL26]. By the beginning of May 2026, we had established the infinite-order boundary vanishing in (1). After some extensive unsuccessful attempts for a few weeks to establish the required unique continuation property (UCP), we decided to write up only the sphericity result in [HL26], which is exactly what is needed for the application in [HHL26]. Professor Huang then suggested me pursuing boundary unique continuation independently in a separate paper. That month, we shared our proposed approach with the free version of ChatGPT and asked for relevant references. Its initial suggestions mainly concerned the Carleman estimates for subelliptic operators and were not particularly useful for our purposes. Also in May, Professor Huang had extensive discussions on this stuff with Professor Sagun Chanillo, a well-known expert on unique continuation for PDE, who recommended through Professor Huang to me starting with a paper by R. Mazzeo [Maz91]. The subsequent careful study beginning with Mazzeo's paper eventually led us to [BH14], where the exact theorem is ready for us to use. In July 2026, out of curiosity, Professor Huang asked me to ask the free version of ChatGPT again how the CSC condition for a Bergman metric implies that it is Kähler–Einstein. To our surprise, its response closely matched the ideas and strategy we had previously supplied and outlined the remaining steps needed to complete the proof.

The whole proofs in [LL26] occupy approximately 7 pages. Much of the argument consists of setting up the problem appropriately, verifying the necessary hypotheses, and applying existing results, with little additional machinery.

I asked ChatGPT-6 the writing style of a main section on UCP in [EG26] (about 14 pages while ours on UCP is about 4 pages) the following question:

Question: Does [EG26]*'s Section 6 have a lot of AI writing style?*

The answer is: *Yes, in a stylistic sense: Section 6 has features often associated with AI-assisted technical writing-extensive signposting, repeated explanations of why each step works, and very explicit bookkeeping of constants, derivatives, cutoffs, and local-to-global estimates.*

As noted in [LL26, Remark 1.3], our argument also establishes the two-dimensional CSC-to-Einstein conclusion under the additional assumption of boundary sphericity. We did not pursue a proof of sphericity in dimension two under the CSC assumption because we had received a preprint by Ebenfelt, Ganguly, and two other authors, intended for posting on arXiv, in which this implication had been proved.

On September 25, 2026, a colleague contacted me to relay a proposal from Ebenfelt, Ganguly, and Zehao Sha to jointly post and publish a version of our paper with five co-authors. Given that our paper had already been posted on arXiv, we did not consider this proposal appropriate and therefore declined.

References

[BH14] O. Biquard and M. Herzlich, *Analyse sur un demi-espace hyperbolique et poly-homogénéité locale*, Calc. Var. Partial Differential Equations **51** (2014), 813–848; arXiv:1002.4106.

[EG26] P. Ebenfelt and S. Ganguly, *Rigidity of Extremal and cscK Bergman Metrics on Pseudoconvex Domains*, arXiv:2610.04590v1, October 3, 2026.

[EX24] P. Ebenfelt and M. Xiao: *Bergman logarithmically flat and obstruction flat hypersurfaces and their CR structures*, Geometric analysis of PDEs and several complex variables, 115–142, Lat. Amer. Math. Ser., Lat. Amer. Math. Ser. UFSCar Subser., Springer, Cham, 2024.

[HHL26] C.-Y. Hsiao, X. Huang and X. Li, *Localization of Bergman Kernels and the Cheng–Yau Conjecture on Real Analytic Pseudoconvex Domains*, arXiv:2604.05542v2, 2026.

[HL26] X. Huang and X. Li, *A note on csc Bergman metric*, arXiv:2606.01319v1, May 31, 2026.

[LL26] X. Li and T. Liu, *CSC Bergman metrics are Einstein on pseudoconvex domains*, arXiv:2609.24853v1, September 21, 2026.

[Ma21] A. H. Martin, *Boundary Asymptotics for Convex and Strongly Pseudoconvex Domains*, Ph.D. thesis, University of California, Riverside, 2021, 111 pp.

[Maz91] R. Mazzeo: Unique continuation at infinity and embedded eigenvalues for asymptotically hyperbolic manifolds, Amer. J. Math. 113 (1991), no. 1, 25–45.

School of Mathematics and Statistics, Wuhan University, Wuhan, Hubei 430072, China.

Email address: xiaoshanli@whu.edu.cn